

JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Combinations, Selection & Multinomial Partitioning	9 PYQs	Selections & Distribution
Moderate	Fundamental Principles & Permutations (Distinct & Identical)	6 PYQs	Arrangements

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Combinations, Selection & Multinomial Partitioning	${}^nC_r = \frac{n!}{r!(n-r)!}$, $\text{Non-negative solutions: } {}^{n+r-1}C_{r-1}$, $\text{Derangements: } D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$
Fundamental Principles & Permutations (Distinct & Identical)	${}^nP_r = \frac{n!}{(n-r)!}$, $\text{Arrangements with duplicates: } \frac{n!}{p! q! r!}$, $\text{Circular: } (n-1)!$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Combinations, Selection & Multinomial Partitioning (High Yield — 9 PYQs)

Core Principle: Combinatorial selection, Pascal identity ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$, distribution of identical items into distinct bins, and derangements.

■ **Caution / JEE Pitfall:** Number of positive integral solutions ($x_i \geq 1$) to $x_1 + x_2 + \dots + x_r = n$ is ${}^{n-1}C_{r-1}$, whereas non-negative ($x_i \geq 0$) is ${}^{n+r-1}C_{r-1}$.

■ **Speed Trick:** Number of derangements of 4 items is $D_4 = 9$; for 5 items is $D_5 = 44$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Stateme..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of four digit even number that can be formed using the digits 0,1,2 and 3 without repetition is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Ten chairs are numbered as 1 to 10. Three women and two men wish to occupy one chair each. First the women choose the chairs marked 1 to 6 ,..." [Ans: A]

3.2 Fundamental Principles & Permutations (Distinct & Identical) (Moderate Yield — 6 PYQs)

Core Principle: Multiplication and addition principles, linear permutations with repetitions, rank of words in dictionary, and circular arrangements.

■ **Caution / JEE Pitfall:** For circular permutations of necklaces/garlands where clockwise and anticlockwise arrangements are indistinguishable, divide by 2: $((n-1)! / 2)$.

■ **Speed Trick:** Gap method: Place items that cannot be adjacent into the gaps created by the remaining unrestricted items.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?..." [Ans: N/A]
- **[2025 Shift] Q#4 (MCQ):** "The number of diagonals that can be drawn in an octagon is..." [Ans: N/A]
- **[2023 Shift] Q#1 (MCQ):** "Ten chairs are numbered as 1 to 10. Three women and two men wish to occupy one chair each. First the women choose the chairs marked 1 to 6 ,..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?

- (A) 720
- (B) 360
- (C) 120
- (D) 180

Correct Answer: None

Solution Analysis: The characters in "KCET 2025" are: K,C,E,T,2,0,2,5 There are 8 characters in total, and the digit 2 is repeated twice. We need arrangements that start with K and end with 5 . So fix K in the first position and 5 in the last position: K_____5 Now we have to arrange the remaining 6 characters: C,E,T,2,0,2 If all were different, the number of arrangements would be: $6!$ But the digit 2 appears twice, so we divide by $2!$: $(6! / 2!) = (720 / 2) = 360$ So the number of ways is: 360 Correct option: Option B .

Benchmark Problem 2: KCET 2026 (Q#2)

10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Statement II: The number of chords that can be formed is 45 Which of the following is correct?

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true and Statement II is false
- (D) Statement I is false and Statement II is true

Correct Answer: None

Solution Analysis: For a triangle, we need to choose 3 points from the 10 distinct points on the circle. Since all points lie on a circle, no three points are collinear. So every selection of 3 points forms a triangle. Number of triangles: ${}^{10}C_3 = (10 \times 9 \times 8 / 3 \times 2 \times 1) = 120$ So Statement I is false, because the number of triangles is not 100 . For a chord, we need to choose 2 points on the circle. Every pair of points gives one chord. Number of chords: ${}^{10}C_2 = (10 \times 9 / 2 \times 1) = 45$ So Statement II is true. Hence, the correct option is: Option D St